

Offline Flyback SMPS — Design Notes (Concise)

System Overview

- Topology: Isolated flyback
- Input: 120 VAC $\rightarrow$ $\sim$ 170 VDC bulk
- Output (prototype): 5.0 V / 0.3 A
- Target PCB: 5.0 V / 1 A
- Controller: TinySwitch (TNY285 class)
- Feedback: TL431 + optocoupler
- Aux winding: None

Primary-Side Design

- EN/UV fed from HV bus via high-value resistor
- Regulation via optocoupler pulling EN/UV
- RCD clamp implemented
- One IC failure without clamp (expected failure mode)

Transformer

- Fully custom hand-wound HF transformer
- Correct polarity/turns on first attempt
- Validated by immediate regulation
- Leakage inferred from ringing

Drain Stress & Clamp Validation

- Measured with 500 $\times$ differential probe
- $V_{\text{bulk}} \approx 170$ VDC
- V_{DS} peak ≈ 340 –345 V
- Large margin to MOSFET rating

Leakage Ringing

- Ringing frequency ≈ 340 –390 kHz
- Ringing amplitude ≈ 100 Vp-p actual
- Ringing decays quickly

Secondary-Side Power Path

- Schottky rectifier for 5 V output
- Initial low capacitance caused ripple/HF spikes

Feedback Loop

- TL431 divider sets 5.00 V
- Divider dissipation in mW range

- ~ 1 nF compensation cap
- Stable from no-load to full load

Load Testing

- $100\ \Omega$ (~ 50 mA)
- $47\ \Omega$ (~ 106 mA)
- $22\ \Omega$ (~ 227 mA)
- ~ 15 – $16\ \Omega$ equiv (~ 300 mA)
- Regulation held at 5.00 V

Output Noise Mitigation

- HF spikes: diode recovery + leakage
- LF ripple: energy packets
- Planned fixes:
 - Secondary RC snubber
 - Increase output capacitance
 - Add μ F-range ceramic
 - Optional π -filter

Efficiency Measurement Plan

- Wall-to-DC efficiency primary metric
- Measure real AC input power
- DC bus power as secondary metric
- Plot efficiency vs load

USB Charger Considerations

- USB-A: BC1.2 DCP (short D+/D–)
- USB-C: 5.1 k Ω pull-downs on CC1 & CC2

Loop Analysis (Planned)

- Middlebrook injection via opto $\rightarrow$ EN/UV
- Sweep ~ 10 Hz to tens of kHz

PCB Readiness

- Electrical design functional first try
- Optional clamp/snubber footprints
- Tight current loops, creepage, test points

Key Outcomes

- First complete assembly regulated immediately
- Custom transformer successful
- Clamp prevents overstress
- Process now repeatable